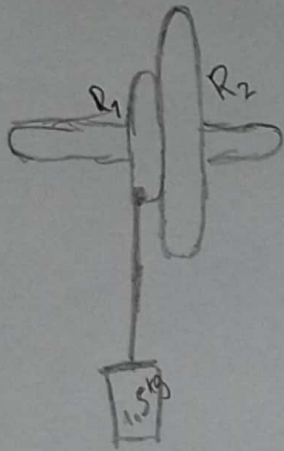


Primer Examen Corto

1) $R_1 = 2.5 \text{ cm}$ $R_2 = 5 \text{ cm}$

$M_1 = 0.8 \text{ kg}$ $M_2 = 7.6 \text{ kg}$



a) $I_T = I_1 + I_2$

$I = \frac{1}{2} MR^2$

$I_T = \frac{1}{2} M_1 R_1^2 + \frac{1}{2} M_2 R_2^2$

$I_T = \frac{1}{2} (0.8 \text{ kg}) (0.025 \text{ m})^2 + \frac{1}{2} (7.6 \text{ kg}) (0.05)^2$

$I_T = 2.25 \times 10^{-3} \text{ kg/m}^2$

$2.5 \text{ cm} \cdot \frac{1 \text{ m}}{100 \text{ cm}} = 0.025$

$5 \text{ cm} \cdot \frac{1}{100 \text{ cm}} = 0.05$

b) $K_i + U_i = K_f + U_f$

$V_2 = 2V_1$

$mgh = \frac{1}{2} mV^2 + \frac{1}{2} I_1 \omega^2 + \frac{1}{2} I_2 \omega^2$

$mgh = \frac{1}{2} mV^2 + \frac{1}{2} \left(\frac{1}{2} MR^2 \right) \frac{V^2}{R^2} + \frac{1}{2} \left(\frac{1}{2} MR^2 \right) \frac{4V^2}{R^2}$

$mgh = \frac{1}{2} mV^2 + \frac{1}{4} MR^2 \cdot \frac{V^2}{R^2} + \frac{1}{4} MR^2 \cdot \frac{4V^2}{R^2}$

$mgh = V^2 \left(\frac{1}{2} m + \frac{1}{4} M + M_2 \right) = \frac{1}{2} V^2 (m + \frac{1}{2} M + 2M) = mgh$

$V = \sqrt{\frac{2mgh}{m + \frac{1}{2} M + 2M}} = \sqrt{\frac{2(1.5 \text{ kg})(9.8 \text{ m/s}^2)(2 \text{ m})}{(1.5 \text{ kg}) + \frac{1}{2}(0.8 \text{ kg}) + 2(7.6 \text{ kg})}} = \underline{\underline{3.4 \text{ m/s}}}$

c) $mgh = \frac{1}{2} mV^2 + \frac{1}{2} I_1 \omega^2 + \frac{1}{2} I_2 \omega^2$

$mgh = \frac{1}{2} mV^2 + \frac{1}{2} \left(\frac{1}{2} MR^2 \right) \frac{V^2}{4R^2} + \frac{1}{2} \left(\frac{1}{2} MR^2 \right) \frac{V^2}{R^2}$

$mgh = \frac{1}{2} mV^2 + \frac{1}{4} MR^2 \cdot \frac{V^2}{4R^2} + \frac{1}{4} MR^2 \cdot \frac{V^2}{R^2} = V^2 \left(\frac{1}{2} m + \frac{1}{4} M + \frac{1}{4} M \right) =$

$V^2 = \sqrt{\frac{2mgh}{m + \frac{1}{8} M + \frac{1}{2} M}} = \sqrt{\frac{2(1.5)(9.8)(2)}{(1.5) + \frac{1}{8}(0.8) + \frac{1}{2}(7.6)}} = \underline{\underline{4.95 \text{ m/s}}}$

2)

$$I = \frac{2}{5} MR^2$$



Ejemplo Sólida

$$I = \frac{2}{3} MR^2$$



Ejemplo hueca

Sólida

$$V = \frac{4}{3} \pi (0.3 \text{ m})^3 = 0.1131 \text{ m}^3$$

$$M = (800 \text{ Kg/m}^3)(0.1131 \text{ m}^3) = \underline{90.48 \text{ Kg}}$$

Hueca

$$M = (20 \text{ Kg/m}^2) \cdot 4\pi \cdot (0.3 \text{ m})^2 = \underline{22.62 \text{ Kg}}$$

$$I = \frac{2}{5} (90.48)(0.3)^2 + \frac{2}{3} (22.62)(0.3)^2 = \underline{6.79 \text{ Kg/m}^2}$$

3) a) $-T = \frac{1}{2} I (\omega^2 - \omega_0^2)$

$$\omega_0 = 300 \frac{\text{rev}}{\text{min}} \cdot \frac{2\pi \text{ rad}}{1 \text{ rev}} \cdot \frac{1 \text{ min}}{60 \text{ s}} = 31.42 \text{ rad/s}$$

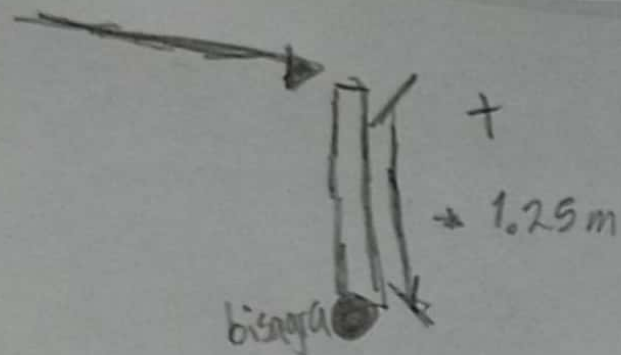
$$\omega = \sqrt{\frac{31.42^2 - \frac{2 \cdot 4000}{76}}{1}} = 22.07 \text{ rad/s}$$

$$\omega = 22.07 \frac{\text{rad}}{\text{s}} \cdot \frac{1 \text{ rev}}{2\pi \text{ rad}} \cdot \frac{60 \text{ s}}{1 \text{ min}} = 210.73 \text{ rpm}$$

b)

$$P = \frac{4000 \text{ J}}{5 \text{ s}} = 800 \text{ W}$$

4)



$$\sum \tau = Fl = (220 \text{ N})(1.25 \text{ m}) = 275 \text{ N}\cdot\text{m}$$

$$I = \frac{1}{3} M l^2 = \frac{1}{3} \left(\frac{750 \text{ N}}{9.8 \text{ m/s}^2} \right) (1.25 \text{ m})^2 =$$

$$I = 39.9 \text{ Kg}\cdot\text{m}^2$$

$$\sum \tau = I \alpha =$$

$$\alpha_z = \frac{\sum \tau_z}{I} = \frac{275}{39.9} = 6.89 \text{ rad/s}^2$$

$$\theta - \theta_0 = 90^\circ \cdot \frac{\pi \text{ rad}}{180^\circ} = \frac{\pi}{2}$$

$$\theta - \theta_0 = \omega_0 t + \frac{1}{2} \alpha_z t^2$$

$$t = \sqrt{\frac{2(\theta - \theta_0)}{\alpha_z}} = \sqrt{\frac{2(\pi/2)}{6.89 \text{ rad/s}^2}} = 0.68 \text{ s}$$

5)

$$V_+ = \sqrt{2gh_s} = \sqrt{2(9.8 \text{ m/s}^2)(12 \text{ m})} = 15.33 \text{ m/s}$$

$$mvd = (5 \text{ kg})(15.33 \text{ m/s})(2 \text{ m}) = 153.3 \text{ kg/m}^2/\text{s}$$

$$I = \frac{1}{12} (8 \text{ kg})(4 \text{ m})^2 = 10.67 \text{ kg/m}^2$$

$$L_f = \left[(15)(2) + \frac{10.67}{2} + (5)(2) \right] V = (25.33 \text{ kg/m}) V$$

$$153.3 \text{ kg/m}^2/\text{s} = (25.33 \text{ kg/m}) V$$

$$E_o = E_p$$

$$\frac{1}{2} mv_o^2 = mgh_s = \frac{1}{2} v^2 = gh_f$$

$$h_f = \frac{v^2}{2g} = \frac{0.054}{2(9.8 \text{ m/s}^2)} = 1.87 \text{ m}$$